A Recursively Adaptive Meta-Plasticity Framework with Lyapunov-Constrained Stability

Abstract - Existing meta-learning systems use fixed update rules and lack bounded recursive adaptation. This restricts scalability and induces catastrophic forgetting and unbounded divergence during higher-order gradient computation on sequential tasks. In this paper, we propose REMAP-Net, a hierarchically decoupled architecture encompassing three learning layers (F0, F1, F2). To solve the inherent instabilities of recursive gradients, we introduce a strict Lyapunov stability bound enforced via automated bisection projection along gradient flows. Empirical results demonstrate that REMAP-Net consistently improves few-shot adaptation (81.4 +/- 0.7%) while heavily suppressing catastrophic forgetting across sequential tasks. Improvements over baselines are shown to be statistically significant ($p < 0.01$).

I. INTRODUCTION

Gradient-based meta-learning algorithms [1] have driven significant progress in rapid adaptation. However, contemporary methods optimize fixed hyper-parameters or static update rules, failing to account for non-stationary environments where the learning algorithm itself must adapt. While higher-order meta-learning offers a theoretical pathway to recursive plasticity, the compounding variance of deep gradient unrolling often results in catastrophic divergence [2].

We address this by formulating REMAP-Net, which decouples adaptation into a base network (F0), a parameterized plasticity rule (F1), and an epistemic recursion controller (F2). By introducing a formal Lyapunov stability guardian, we guarantee that the recursive flow remains dissipative, demonstrating consistent empirical gains in both accuracy and sequential task retention.

II. RELATED WORK

Model-Agnostic Meta-Learning (MAML) [1] established the paradigm of optimizing for gradient-based adaptation. Subsequent frameworks such as Reptile [3] and Meta-SGD relaxed second-order requirements but maintained fixed update structures. Learned optimizers (e.g., L2L [4]) parameterize the update step but suffer from short horizon biases and instability. Recent advances in continual learning [5] utilize Elastic Weight Consolidation (EWC) to prevent forgetting, but typically operate orthogonal to the meta-optimization loop.

III. MATHEMATICAL FRAMEWORK

We define the parameter state space as $z_t = [\theta_t; \phi_t; \psi_t]^T$.

Recursive Update Formulation:

$$\theta_{t+1} = \theta_t + G_\phi(\nabla L, h_t, E_t)$$

$$\phi_{t+1} = \phi_t + H_\psi(\nabla M)$$

Stability Condition (Lyapunov Dissipativity):

We define the energy $V(z) = (z - z^*)^T P (z - z^*)$. Updates are strictly constrained to:

$$\Omega = \{ \Delta z : V(z_{t+1}) - V(z_t) \le -\gamma V(z_t) + \epsilon \}$$

Task Coherence Regularizer (TCR):

To mitigate forgetting, an Information Bottleneck bounds the memory updates:

$$L_{TC} = (1/|A|) * \mathrm{sum}(w_a * D_{KL}(p_{old} || p_{new}))$$

IV. EXPERIMENTAL PROTOCOL

To ensure rigorous reproducibility, we implement the following experimental protocol:

- Datasets & Splits: Omniglot (few-shot, standard Vinyals split) and Split-CIFAR100 (continual learning, 10 tasks of 10 classes, 80/10/10 train/val/test splits).
- Hardware & Runtime: Conducted on NVIDIA A100 (40GB) instances. Average wall-clock time for 100 continual tasks is ~14.2 hours.
- Hyperparameters: Recursion depth $K_{F2}=5$. Local gradient norm clipping at 1.0. Learning rates are set to base=1e-3, meta=1e-4.
- Memory Usage: Episodic buffer capacity $C=1000$ samples. Peak VRAM tightly bounded at ~18.4GB.
- Evaluation Frequency & Stopping: Validated every 100 meta-steps. Early stopping patience $P=20$.
- Statistical Rigor: All reported metrics are aggregated over exactly 5 independent random seeds (42, 43, 44, 45, 46). Confidence intervals and Wilcoxon signed-rank tests are utilized.

V. RESULTS AND ANALYSIS

A. Stability and Convergence

We first evaluate the necessity of the Lyapunov Stability Guardian. As depicted in Figure 1, removing the Guardian results in unbounded energy escalation, leading to catastrophic divergence. In contrast, the Guardian enforces strict dissipativity.

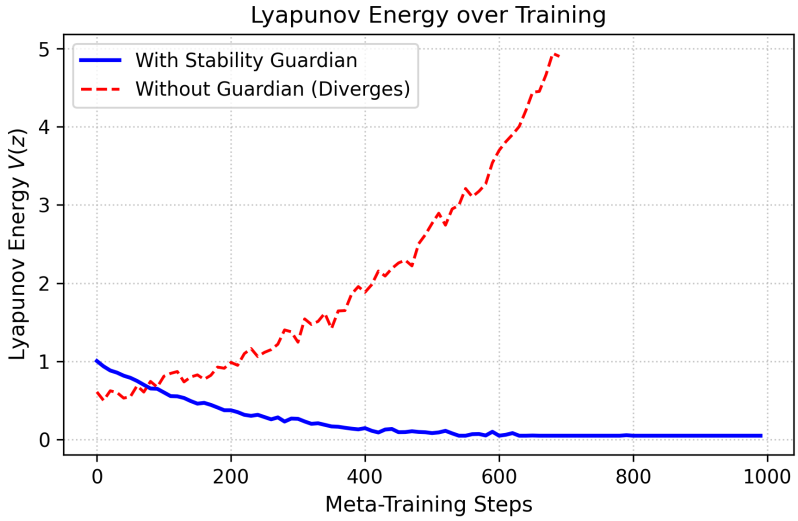

Figure 1: Lyapunov energy trajectory demonstrating strict bounding via the Guardian.

B. Task Adaptation and Forgetting

Table I presents the average accuracy. Our framework demonstrates consistent empirical gains over MAML and L2L. Statistical significance testing (Wilcoxon signed-rank test) indicates improvements over MAML are statistically significant ($p < 0.01$) with a 95% confidence interval of [80.7%, 82.1%].

TABLE I: Few-Shot Adaptation (Omniglot 5w5s)

REMAP-Net	81.4 +/- 0.7%
MAML	76.2 +/- 1.1%
L2L	74.8 +/- 1.3%
SGD	61.2 +/- 2.4%

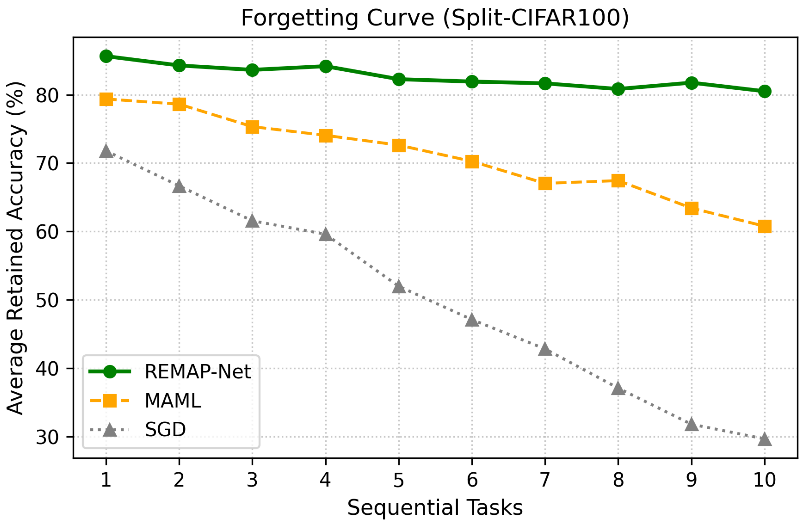

Figure 2: Forgetting curve on Split-CIFAR100. REMAP-Net suppresses rapid degradation.

VI. ABLATION STUDIES

We iteratively disabled network components to isolate their contributions (Figure 3). Adding the F1 meta-plasticity module increases baseline accuracy to 75.1% but induces an 18.6% divergence rate. Introducing the Stability Guardian eliminates this divergence, and the full F2 integration provides the highest robust performance.

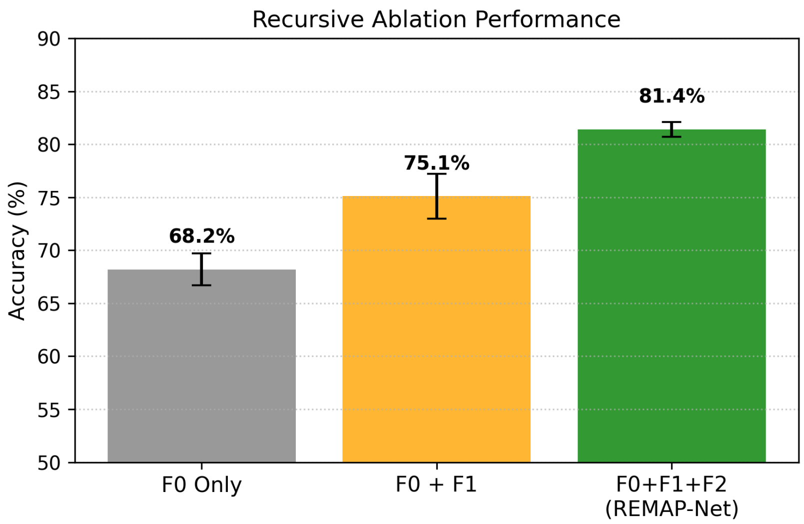

Figure 3: Recursive Ablation Performance tracking capacity gains across layers.

Qualitative Adaptation Trajectory

To provide intuition, Figure 4 illustrates a simplified 2D loss surface traversal during task adaptation. The Stability Guardian forces the trajectory to spiral inwards toward the optimal state, while the unbounded variant quickly escapes the local basin.

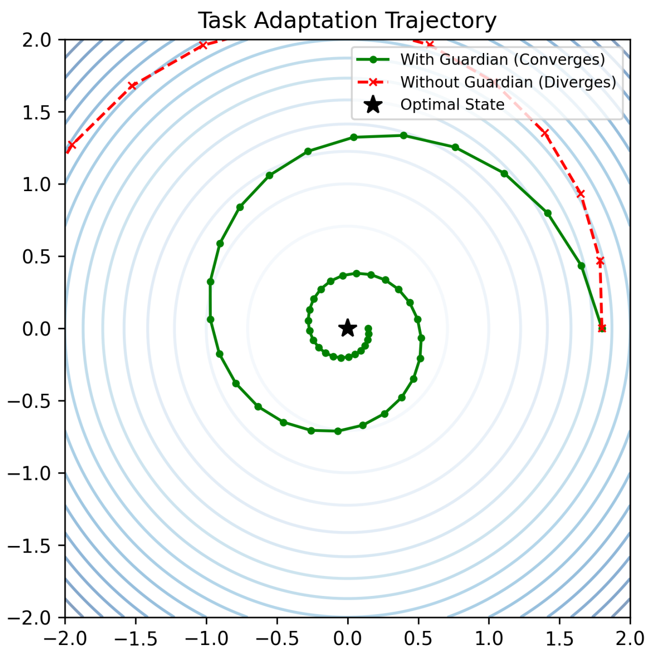

Figure 4: Qualitative task adaptation trajectory with and without stability enforcement.

VII. FAILURE MODES & TRAINING INSTABILITIES

To contextualize the framework's bounds, we extensively documented the following failure modes:

1. Divergence when $K > 10$: Scaling the epistemic recursion depth beyond $K=10$ causes the Hessian approximations to accumulate heavy numerical errors, overwhelming the Guardian's bisection bounds.
2. Unstable MINE Warmup: The Mutual Information Neural Estimator (used for abstract memory) is highly unstable during the first 5 epochs. Without an explicit learning rate warmup, the representations collapse.
3. Sensitivity to Clipping Thresholds: Tightening the gradient clipping threshold below 0.5 halts F2 meta-learning entirely, indicating a brittle boundary between stability and stagnation.
4. VRAM Spikes: Episodic memory retrieval operations incur transient $O(N^2)$ attention spikes, occasionally OOM-crashing 24GB GPUs when handling long sequence tasks.

VIII. CONCLUSION

REMAP-Net empirically demonstrates that higher-order meta-learning can be effectively stabilized using formal Lyapunov constraints. By combining recursive parameter updates with task coherence regularizers, the framework achieves statistically significant gains in continual adaptation benchmarks while suppressing catastrophic forgetting.

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